

List of Formulae

Exponential & Logarithmic Functions

The average rate of change of the function $f(x)$ on the interval $[c, d]$ is: $\frac{f(d)-f(c)}{d-c}$

Compound interest Formula $A = P \left(1 + \frac{r}{n}\right)^{n \cdot t}$

Continuous compound Interest Formula $A = Pe^{r \cdot t}$

$a^x = b \Leftrightarrow x = \log_a b$, where $a > 0, b > 0, a \neq 1$

$\log_b mn = \log_b m + \log_b n$

$\log_b \frac{m}{n} = \log_b m - \log_b n$

$\log_b m^n = n \log_b m$

$\log_b m = \frac{\log_c m}{\log_c b}$

Statistics

Mean (μ) = $\frac{\sum x}{n}$

Standard deviation (σ) = $\sqrt{\frac{\sum (x-\mu)^2}{n}}$

Z-value for a data: $Z = \frac{X-\mu}{\sigma}$

Empirical rule

